



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CANDY MATCH SAGA: CANDY AI LAB

An Intelligent Rule-Based Candy Game with Player Analytics

“Where Every Match Teaches AI Something New.”

CSA0920 – Programming in Java

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QUICK RECAP

Project Concept:

- Candy Match Saga is an intelligent rule-based candy matching game.
- Players swap candies to create valid matches and earn points.
- The game supports Match 2, Match 3, Match 4 and Match 5 combinations.
- Custom candy synthesis and game rules can be created dynamically.
- Player performance is monitored using score, moves, matches and analytics.

Five Planned Modules:

1. Game Setup & Level Management
2. Candy Board & Match Processing
3. Candy Synthesis & Rule Management
4. AI Opponent & Adaptive Strategy
5. Player Analytics & Data Management

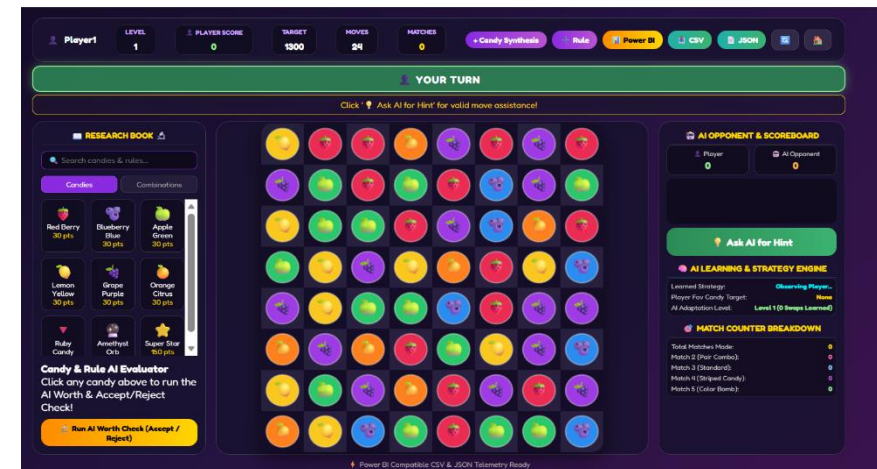
IMPLEMENTATION STATUS SUMMARY



Modules	Status	Approx. Completion
Game Setup & Level Management	Completed	100%
Candy Board & Match Processing	Completed	90%
Candy Synthesis & Rule Management	Completed	90%
AI Opponent & Adaptive Strategy	In Progress	85%
Player Analytics & Data Management	In Progress	80%

Overall Project Completion: ~89%

Status Note: Core gameplay and major dashboard features are functional. Remaining work mainly involves final testing, database integration if required, and report/documentation completion.



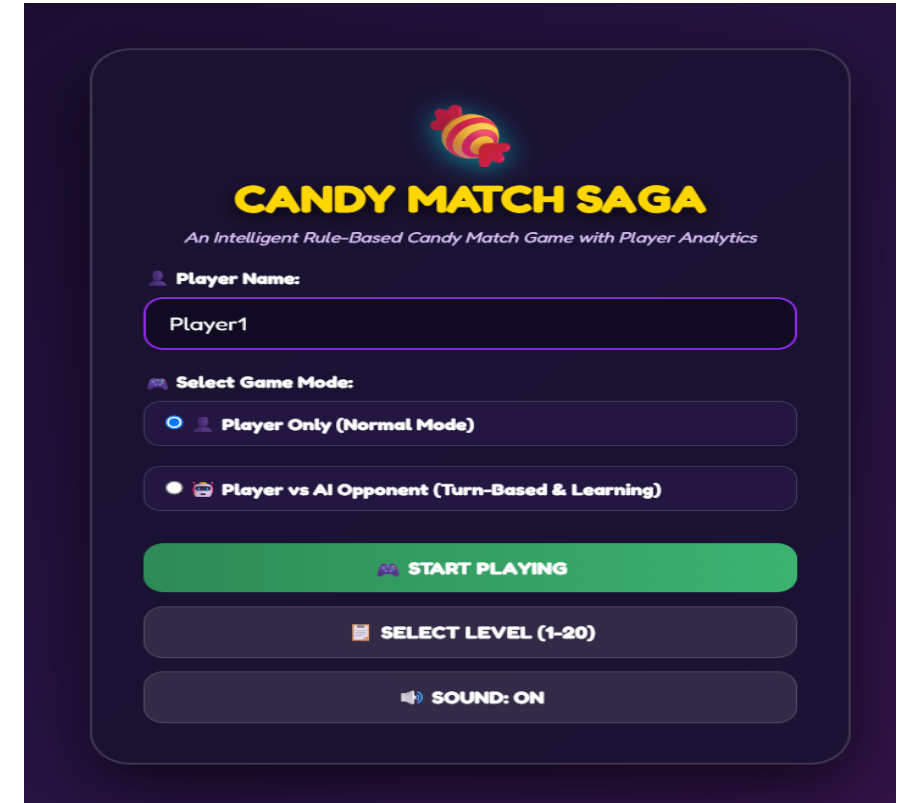
MODULE 1: Game Setup & Level Management

Implemented Features:

- Player name can be entered before starting the game.
- Supports Player Only and Player vs AI modes.
- Players can select levels from 1–20.
- Sound can be enabled or disabled.
- Initializes the game according to the selected player and level.

Key UI:

Player Name → Game Mode → Start Playing →
Select Level → Sound



MODULE 2: Candy Board & Match Processing

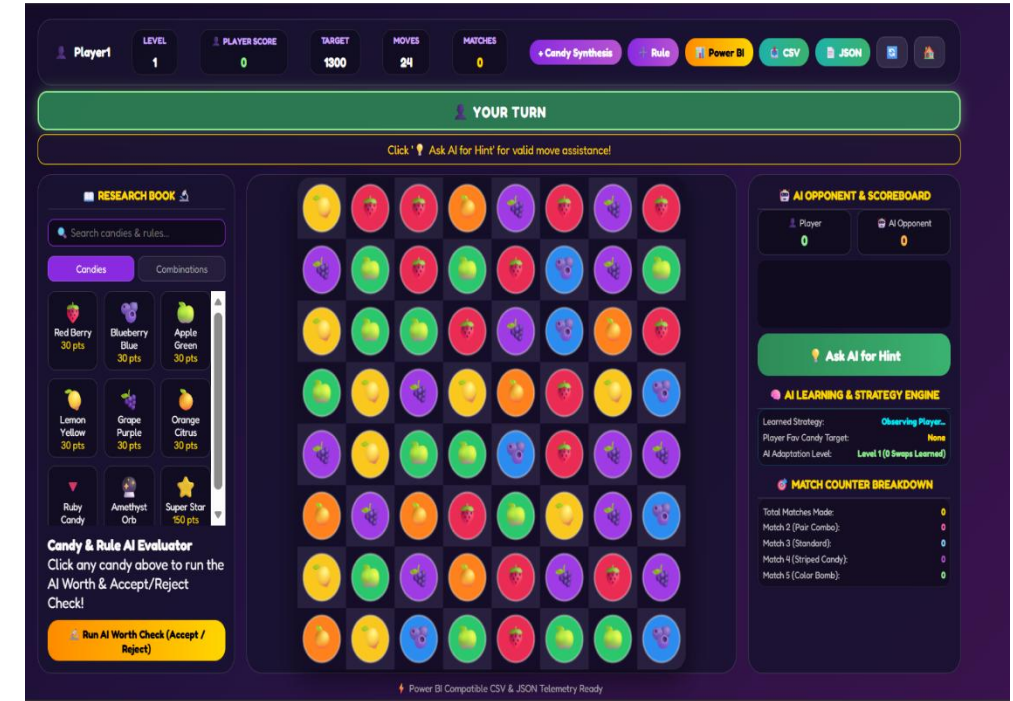


Implemented Features:

- Displays a structured candy grid for gameplay.
- Supports multiple candy types such as Red Berry, Blueberry, Apple, Lemon, Grape and Orange.
- Detects valid candy combinations.
- Supports Match 2, Match 3, Match 4 and Match 5.
- Updates score, moves, target and match counters during gameplay.

Example:

Match 2 → Pair Combo
Match 3 → Standard Match
Match 4 → Striped Candy
Match 5 → Color Bomb



MODULE 3: Candy Synthesis & Rule Management



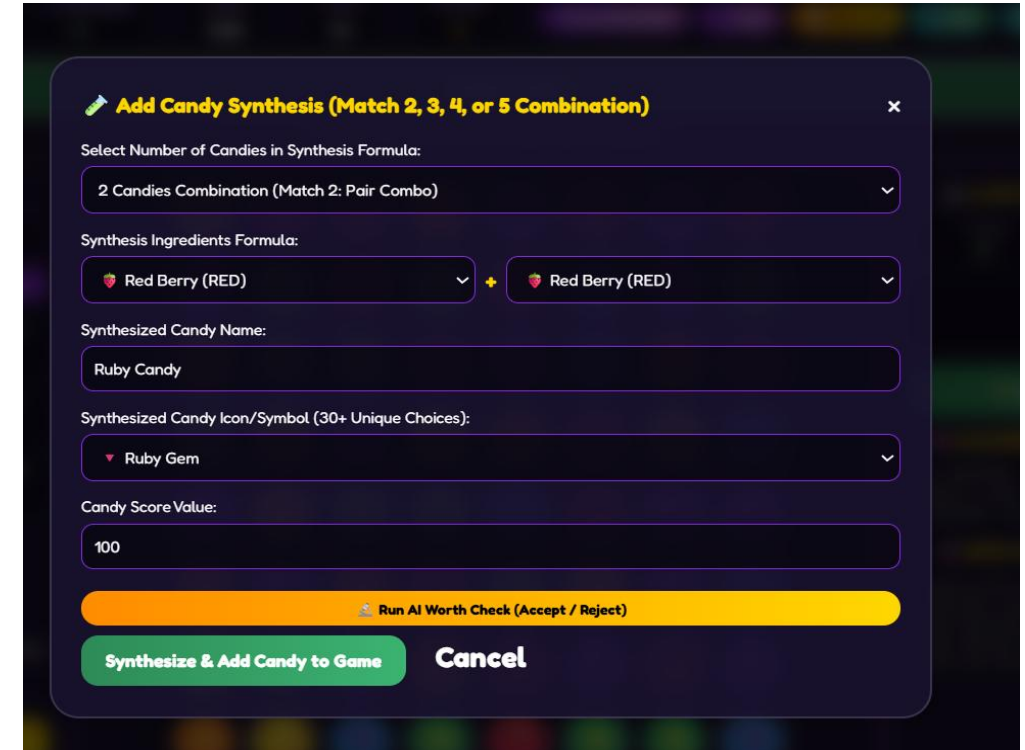
Implemented Features:

- Allows synthesis using 2, 3, 4 or 5 candies.
- User can select ingredients for a synthesis formula.
- Allows creation of custom synthesized candy names and symbols.
- Supports custom game rules such as Ruby Combination Bonus.
- AI Worth Check evaluates whether a candy/rule should be accepted or rejected.

Example:

Red Berry + Red Berry → Ruby Candy (+100 pts)

Lemon + Red Berry + Ruby Candy → Super Star (+150 pts)



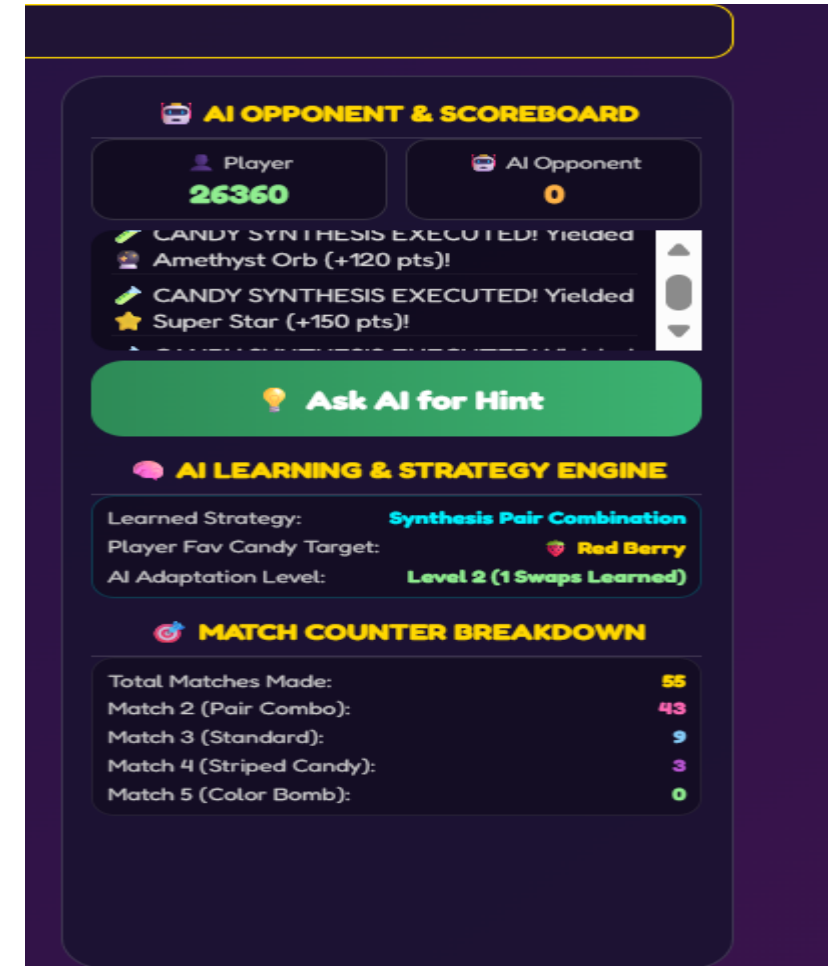
MODULE 4: AI Opponent & Adaptive Strategy

Implemented Features:

- Provides **AI-based hints** for valid player moves.
- Supports **Player vs AI Opponent** mode.
- Tracks player candy preferences and gameplay behaviour.
- Learns successful player strategies.
- Dynamically adapts the AI level based on observed player actions.

AI Learning & Strategy Engine:

- Learned Strategy: Synthesis Pair Combination
- Player Favourite Target: Red Berry
- AI Adaptation Level: Level 2 – Swaps Learned



MODULE 5: Player Analytics & Data Management

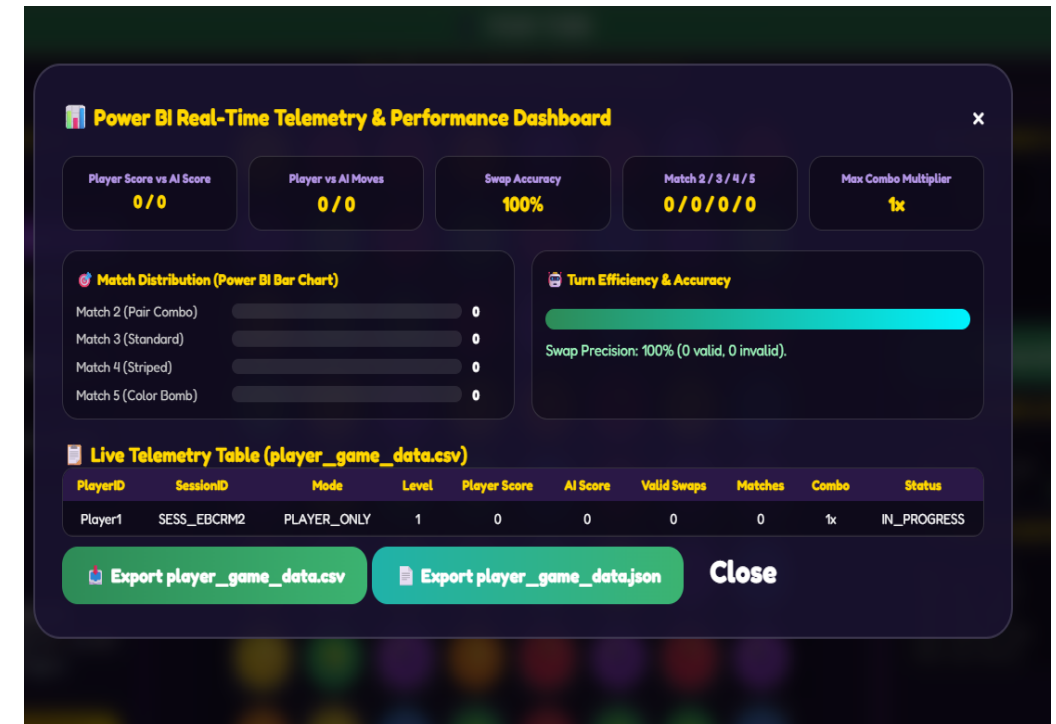


Implemented Features:

- Tracks player score and AI score.
- Records valid swaps and total matches.
- Categorizes matches into Match 2, 3, 4 and 5.
- Provides real-time performance visualization through the dashboard.
- Supports **CSV and JSON data export**.

Analytics Displayed:

- Player Score vs AI Score
- Player vs AI Moves
- Swap Accuracy
- Match Distribution
- Combo Multiplier
- Match Counter Breakdown



COLLECTION FRAMEWORK USAGE

Collection	Usage in Game
ArrayList / List	Store match history and game events
HashMap / Map	Store candy names, symbols and score values
Array / 2D Array	Represent the candy game board
List	Store valid moves / matched candies
Map	Store player statistics and analytics data

Why Collections?

Collections provide efficient storage and retrieval of dynamic game data such as candies, matches, player statistics and rules.

MULTITHREADING IMPLEMENTATION

- Background game tasks can be executed independently without blocking the main game flow.
- Timer and periodic game updates can run separately from player interactions.
- AI decision-making can be processed independently when required.
- Background telemetry/data updates can be handled without interrupting gameplay.
- This improves game responsiveness and provides smoother user interaction.

EXCEPTION HANDLING IMPLEMENTATION

Exception Handling:

- Input validation is performed for player and game configuration.
- Invalid candy/rule combinations can be handled safely.
- Invalid numerical values are prevented from causing application failure.
- Data export errors can be handled using exception handling.
- try-catch blocks prevent unexpected errors from crashing the application.



JDBC / DATABASE INTEGRATION STATUS



Status: In Progress

- Database structure is being finalized for player and game-session data.
- JDBC will be used to connect the game application with MySQL.
- Player details, level, score and gameplay statistics will be stored.
- Match history and performance data can be retrieved for analytics.
- Database integration will provide persistent storage for player records.

LIVE DEMO

1. Start Game
- ↓
2. Select Game Mode & Level
- ↓
3. Perform Candy Swaps
- ↓
4. Generate Matches
- ↓
5. Synthesize Special Candy
- ↓
6. Use AI Hint / AI Opponent
- ↓
7. View Player Analytics
- ↓
8. Export CSV / JSON Data

ISSUES FACED & HOW RESOLVED

Issue 1 – Match Detection

Problem: Multiple match types required different validation conditions.

Solution: Implemented separate rule conditions for Match 2, Match 3, Match 4 and Match 5.

Issue 2 – Dynamic Candy Synthesis

Problem: Different ingredient combinations produce different special candies.

Solution: Created a configurable synthesis and rule system instead of hard-coding every combination.

Issue 3 – AI Adaptation

Problem: AI needed to respond to changing player behaviour.

Solution: Player moves, favourite candy targets and successful strategies are tracked for adaptive decision-making.

DEVIATIONS FROM REVIEW 1 PLAN

Changes Made:

- Added Candy Synthesis as an interactive feature.
- Added Custom Game Rule creation.
- Added AI Worth Check for candies and rules.
- Added AI Learning & Strategy Engine.
- Added Power BI-style real-time telemetry dashboard.
- Added CSV and JSON export for gameplay analytics.

Reason:

The scope was extended to make the game more intelligent, configurable and analytics-oriented.

REMAINING WORK PLAN TO REVIEW 3



Task	Status
Complete AI strategy refinement	In Progress
Complete JDBC/MySQL integration	In Progress
Final exception handling	In Progress
Complete test cases	In Progress
Validate all synthesis combinations	In Progress
Final UI testing	In Progress
Capture final screenshots	Pending
Complete final PPT & Report	Pending

Target:

Complete all functional modules → Testing → Database validation → Final documentation

TESTING NOTES SO FAR



Test Case	Expected Result
Valid candy swap	Match detected and score updated
Invalid swap	Move rejected / no valid match
Match 2	Pair combo generated
Match 3	Standard candy match generated
Match 4	Special/striped candy generated
Match 5	Color bomb generated
Candy synthesis	New candy created
Custom rule	Rule saved and applied
AI hint	Valid move suggestion displayed
CSV export	Gameplay data exported successfully
JSON export	Gameplay data exported successfully



Engineer to Excel



THANK YOU